

SkillSphere - Student Skill Exchange & Peer Learning Platform

Hackathon Blueprint (24-Hour Development Plan)

Executive Summary: SkillSphere digitizes peer learning by enabling students to discover mentors, exchange skills, schedule sessions, verify attendance, earn certificates and badges, while giving administrators analytics and role-based management.

Detailed Features

- Role-based authentication (Admin/Staff/Mentor/Student)
- Skill directory & search
- Skill request portal
- Session booking & calendar
- QR/OTP attendance
- Real-time notifications
- Announcements
- Feedback & ratings
- Certificates (PDF)
- Leaderboards & badges
- Analytics dashboard
- Department reports
- Profile management
- Cloud image uploads

Functional Requirements

- User registration/login
- Manage profiles
- CRUD skills
- Search/filter mentors
- Book/accept/reject sessions
- Attendance verification
- Feedback
- Notification delivery
- Admin analytics
- Department management
- Certificate generation
- Report export

Problem → Solution Mapping

Manual Process	Existing Problem	SkillsSphere Solution
Finding skilled students	Word of mouth	Searchable skill directory
Asking for help	WhatsApp messages	Skill request portal
Session scheduling	Manual coordination	Calendar booking
Attendance	Paper sheet	QR/OTP verification
Notifications	Manual reminders	Automated notifications
Certificates	Manual	Auto-generated certificates
Skill verification	Trust-based	Faculty approval
Mentor recognition	None	Badges & leaderboard
Reports	Excel	Analytics dashboard
Department tracking	Impossible	Department reports
Feedback	Google Forms	Built-in feedback

Non-Functional Requirements

- Support 15,000+ users
- <2 second response time
- JWT & bcrypt security
- RBAC
- HTTPS
- MongoDB indexing
- Responsive UI
- Scalable REST APIs
- Logging & backups
- 99.9% availability target

Technology Stack

Layer	Technology
Frontend	React (Vite), Tailwind CSS
Backend	Node.js + Express
Database	MongoDB Atlas
ODM	Mongoose
Authentication	JWT + bcrypt
Realtime	Socket.IO
Storage	Cloudinary
Charts	Recharts
Email	Nodemailer

PDF	PDFKit
Deployment	Vercel + Render + MongoDB Atlas

24-Hour Execution Plan

- 0-2h: Requirements, UI, DB schema
- 2-6h: Auth + Backend APIs
- 6-10h: Skills & Profiles
- 10-14h: Sessions + Attendance
- 14-18h: Notifications + Certificates
- 18-21h: Analytics + Reports
- 21-24h: Testing, Deployment, Demo